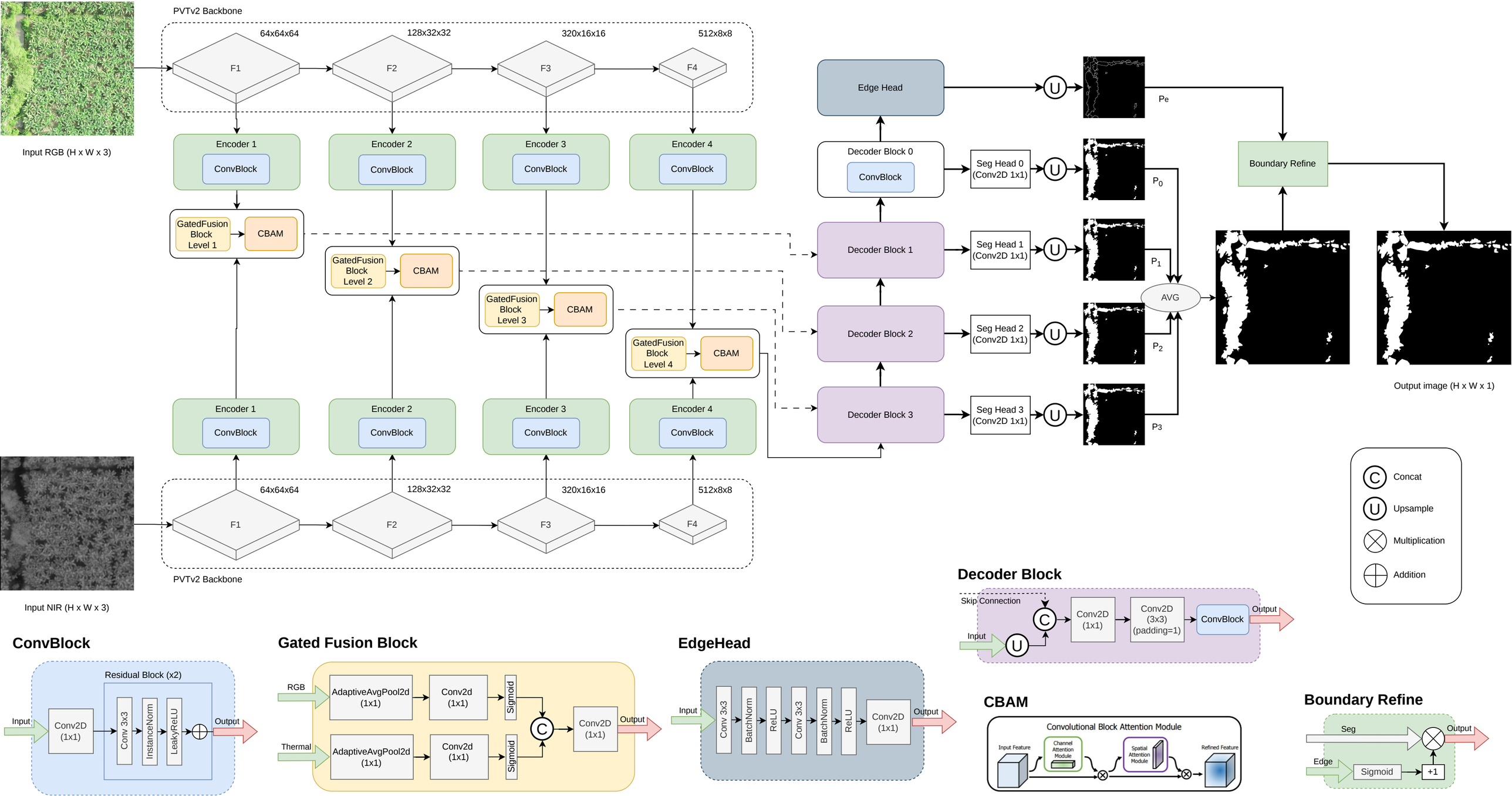
PVTv2 Backbone
64x64x64
128x32x32
320x16x16
512x8x8
F1
F2
F3
F4
Input RGB (H x W x 3)
Input NIR (H x W x 3)
Encoder 1
Encoder 2
Encoder 3
Encoder 4
ConvBlock
GatedFusion Block Level 1
GatedFusion Block Level 2
GatedFusion Block Level 3
GatedFusion Block Level 4
CBAM
Edge Head
Decoder Block 0
Decoder Block 1
Decoder Block 2
Decoder Block 3
Seg Head 0 (Conv2D 1x1)
Seg Head 1 (Conv2D 1x1)
Seg Head 2 (Conv2D 1x1)
Seg Head 3 (Conv2D 1x1)
U
Pe
P0
P1
P2
P3
AVG
Boundary Refine
Output image (H x W x 1)
Concat
Upsample
Multiplication
Addition
Decoder Block
Skip Connection
Input
Conv2D (1x1)
Conv2D (3x3) (padding=1)
Output
ConvBlock
Residual Block (x2)
Conv 3x3
InstanceNorm
LeakyReLU
Gated Fusion Block
RGB
Thermal
AdaptiveAvgPool2d (1x1)
Conv2d (1x1)
Sigmoid
EdgeHead
BatchNorm
ReLU
CBAM
Convolutional Block Attention Module
Input Feature
Channel Attention Module
Spatial Attention Module
Refined Feature
Boundary Refine
Seg
Edge
+1